

OceanBestPractices System:

a global resource to facilitate harmonizing
practices in ocean observation, data and information

Best Practices Working Group:

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IMDIS 2018 – Barcelona (SPAIN), 5-7 November



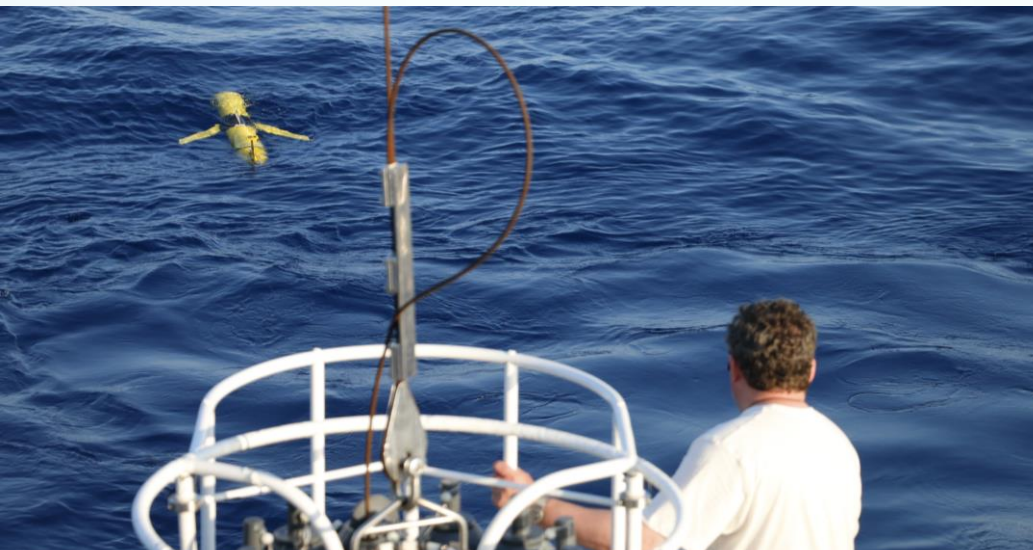
Best Practice Vision

- To have agreed methods for every activity in ocean observing research, operations and applications that are broadly adopted.



Best Practice Mission

- To provide and sustain a system which fosters collaboration, consensus building, and innovation by providing coordinated and global access to best practices across ocean sciences.



Challenges of Best Practices

- Quality of BP Documentation varies widely
- Data and metadata formats are inconsistent
- Machine readability is limited (if at all)
- Sustainability is often not guaranteed
- Lack of coordination/awareness among projects and organizations



Ocean Best Practices System

- Living, Sustained, Comprehensive System for Ocean Observing Best Practices
- Academic recognition through peer review and DOIs
- Improved visibility from search engines
- Training resource
- Greater Interoperability across programs and institutions

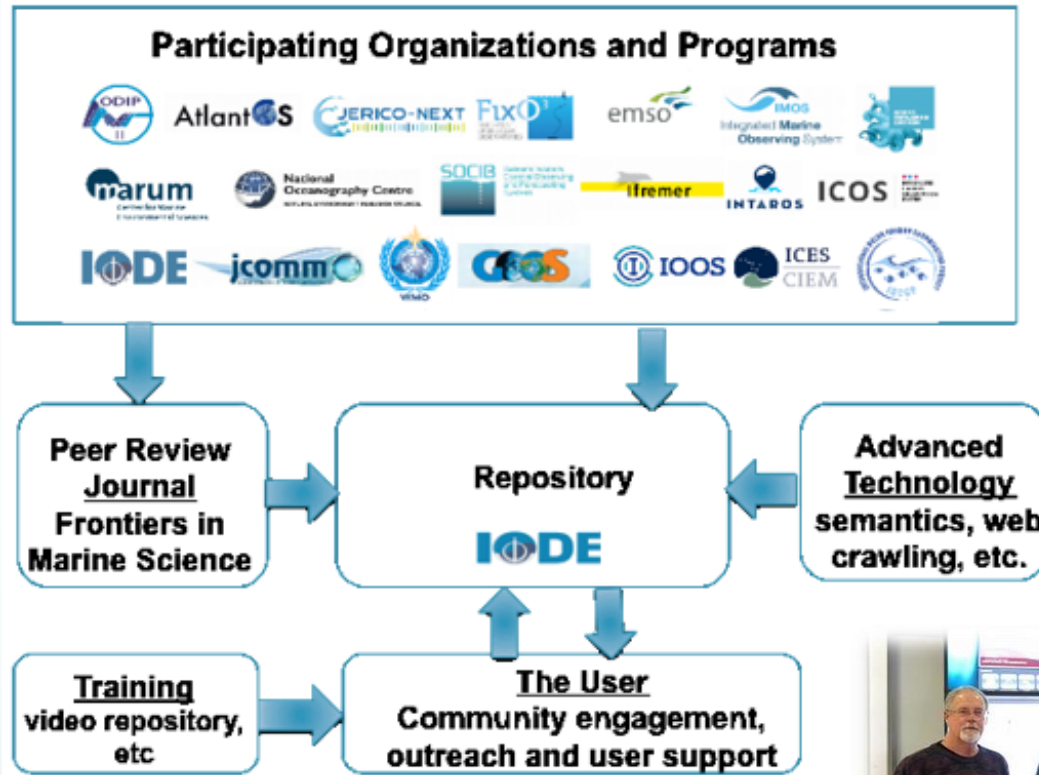


Benefits for the Ocean Observing Community

- **Open Access** to documented BPs
- **A Sustained Repository** to host BPs
- **Peer reviewed publications** linked to BPs
- **Community service** update & acknowledgement of skills and BPs quality



Addressing such challenges





**Societal
Requirements**

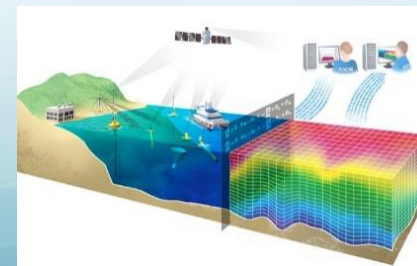
**Scientific
Approach**



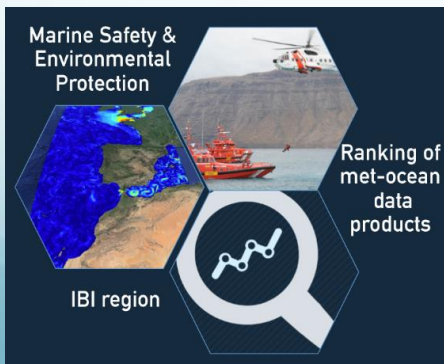
**Data
Acquisition**



**Data
Management
Services**



**Products and
Services**



**Societal
Benefits**



Evolving and Sustaining Ocean Best Practices Workshop, IOC, Paris, France, Nov 2017

“A community best practice is a methodology that has repeatedly produced superior results relative to other methodologies with the same objective.”

To be fully elevated to a best practice, a promising method needs to be adopted and employed by multiple organizations.”

EOS article: Citation: Hermes, J., J. Pearlman, and P. L. Buttigieg (2018), What's the best way to responsibly collect ocean data?, Eos, 99, <https://doi.org/10.1029/2018EO096533>. Published on 04 May 2018.

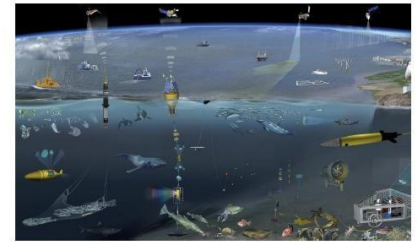


Image credit: NEXOS Project

Evolving and Sustaining Ocean Best Practices Workshop

15 – 17 November 2017
Intergovernmental Oceanographic Commission, Paris, France

Proceedings

Editors

Pauline Simpson, Francoise Pearlman and Jay Pearlman

Rapporteurs:

Mark Bushnell, Juliet Hermes, Cristian Munoz

January 2018

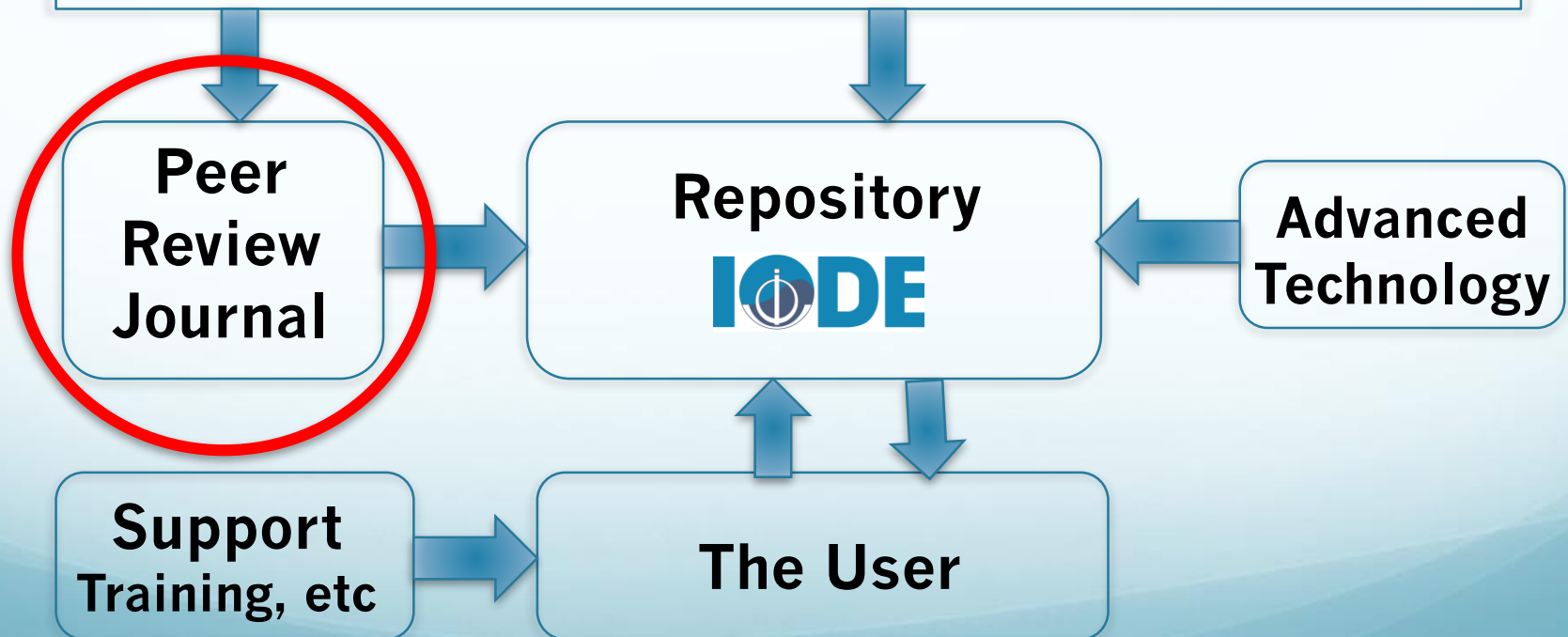


FAIR Principles adopted in OBP-S

**BPs meet domain-relevant
community methodologies??**



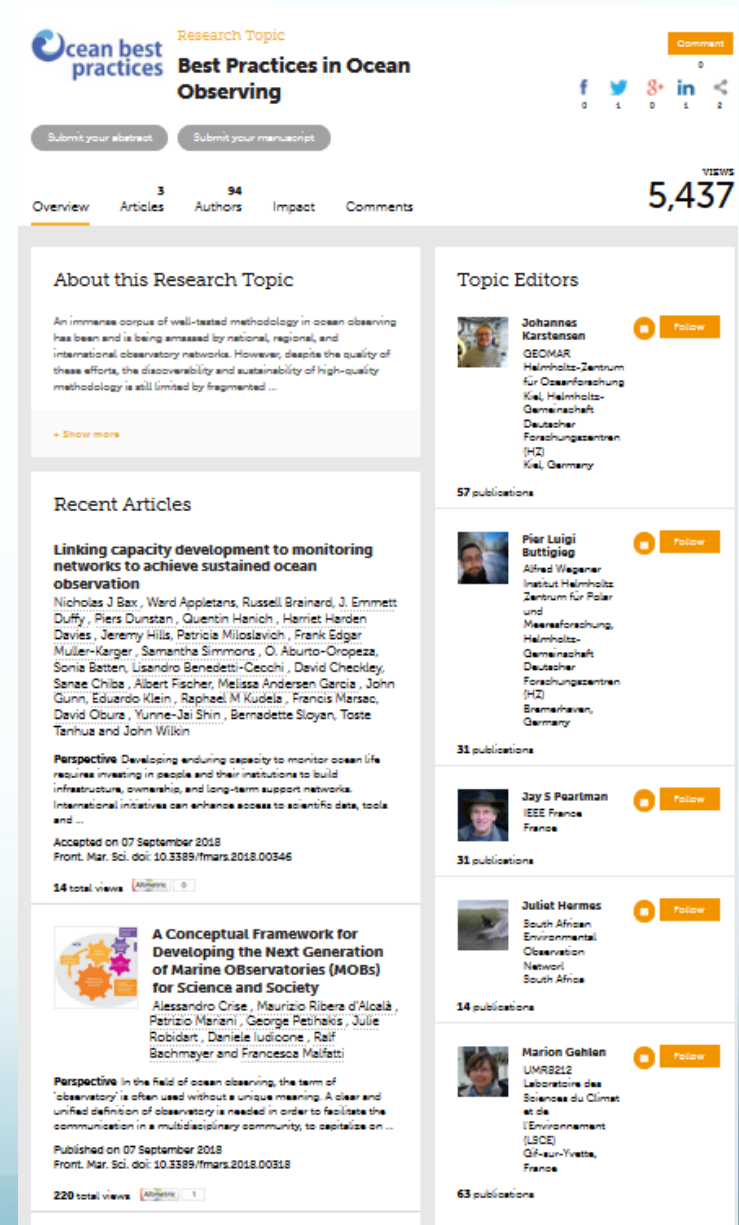
Participating Organizations and Programs





Best Practices in Ocean Observing

- Open access
- Open process review
- Linked to the OBP-R



Research Topic
Best Practices in Ocean Observing

Submit your abstract Submit your manuscript

Overview **3** Articles **94** Authors Impact Comments

5,437 VIEWS

About this Research Topic

An immense corpus of well-tested methodology in ocean observing has been and is being amassed by national, regional, and international observatory networks. However, despite the quality of these efforts, the discoverability and sustainability of high-quality methodology is still limited by fragmented ...

[+ Show more](#)

Recent Articles

Linking capacity development to monitoring networks to achieve sustained ocean observation

Nicholas J Bax, Ward Appeltans, Russell Brainerd, J. Emmett Duffy, Piers Dunstan, Quentin Hanich, Harriet Harden Davies, Jeremy Hills, Patricia Miloslavich, Frank Edgar Muller-Karger, Samantha Simmons, O. Aburto-Oropeza, Sonie Betten, Lisandro Benedetti-Cecchi, David Checkley, Senze Chiba, Albert Fischer, Melissa Andersen Garcia, John Gunn, Eduardo Klein, Raphael M Kudela, Francis Marso, David Obuze, Yunne-Jai Shin, Bernadette Sloyan, Toste Tanhua and John Wilkin

Perspective Developing enduring capacity to monitor ocean life requires investing in people and their institutions to build infrastructure, ownership, and long-term support networks. International initiatives can enhance access to scientific data, tools and ...

Accepted on 07 September 2018
Front. Mar. Sci. doi: 10.3389/fmars.2018.00346

14 total views 

A Conceptual Framework for Developing the Next Generation of Marine Observatories (MOBs) for Science and Society

Alessandro Crise, Maurizio Ribera d'Alcalá, Patrizio Mariani, George Petihakis, Julie Robidart, Daniele Iudicone, Raif Bachmeyer and Francesca Malfatti

Perspective In the field of ocean observing, the term of 'observatory' is often used without a unique meaning. A clear and unified definition of observatory is needed in order to facilitate the communication in a multidisciplinary community, to capitalize on ...

Published on 07 September 2018
Front. Mar. Sci. doi: 10.3389/fmars.2018.00318

220 total views 

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14 publications

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UMR8212
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Sciences du Climat
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l'Environnement
(LSE)
Orfèvre, France

63 publications

Ocean Best Practices System

Participating Organizations and Programs



Peer
Review
Journal

Repository
IODE

Advanced
Technology

Support
Training, etc

The User

Community Engagement



Using BPs

Training & Support

**Reviewing &
Publishing**

**Documenting &
Disseminating**

WAY FORWARD

- Long-term “Home” for the OBPS
- “Inclusive” – community based
 - IOC of UNESCO
 - E-Repository and System
hosted/managed by
IOC Project Office
for IODE
 - IT support offered



OBPS: IOC project

- IODE and GOOS will jointly request establishment of **OBPS Project** by 30th Session of IOC Assembly (26 June – 4 July 2019)
- IOC Steering Group for OBPS will be established
 - Project manager/chief editor
 - Technical manager
 - Representatives of partner communities
 - ...

OBPS and ODIS

- **ODIS = IOC Ocean Data and Information System**
- *“to improve the accessibility and interoperability of existing data and information, and to contribute to the development of a global ocean data and information system, to be referred to as the IOC Ocean Data and Information System, leveraging established solutions where possible”.*

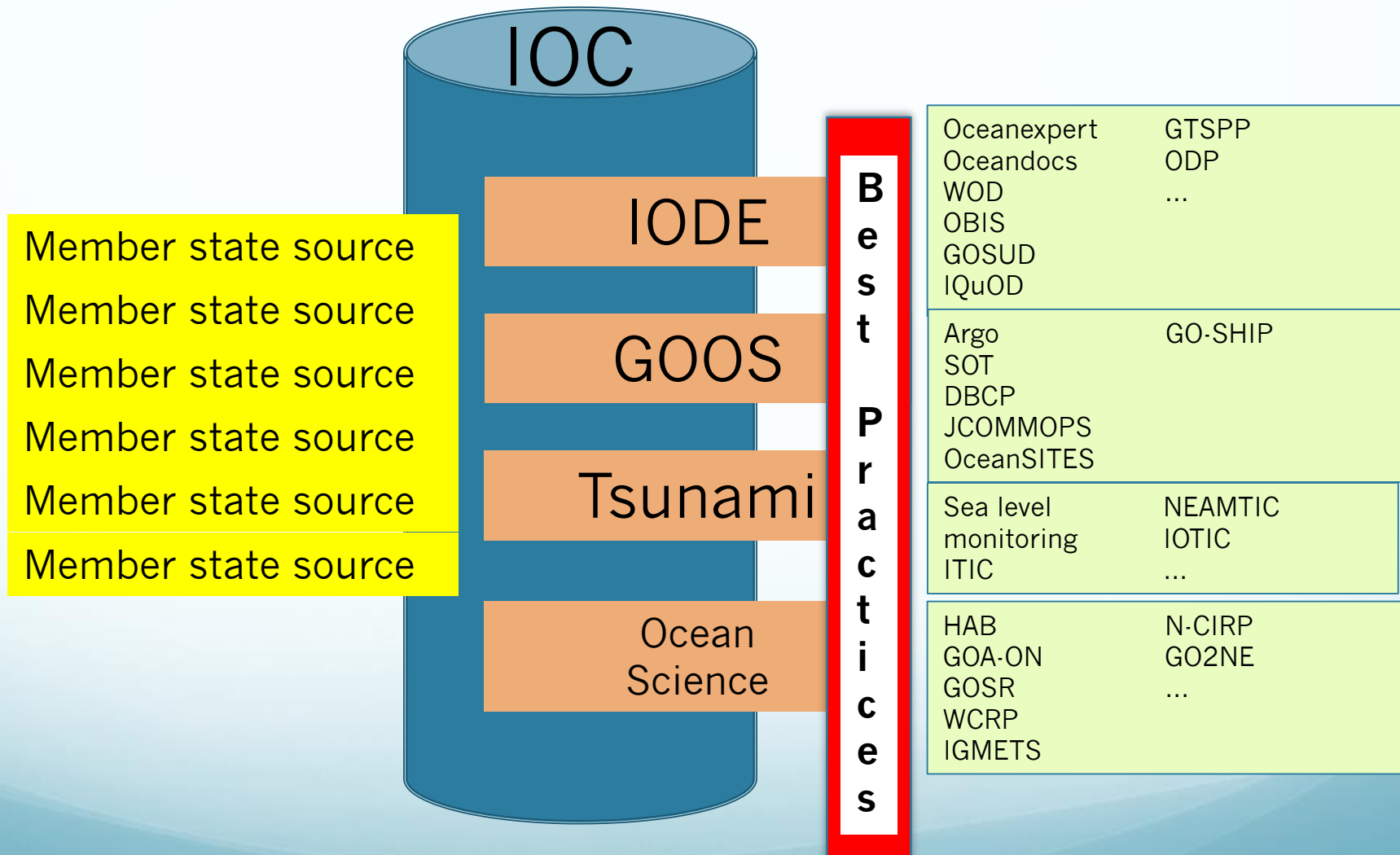
Building on existing efforts

Examples of the types of efforts we hope to leverage:

- Knowledge platform efforts
- Thematic networks
- Metadata and data brokering/exchange
- Semantics and other foundation activities
- Platform development
- **Sharing of best practices**



ODIS : "*shop*" concept



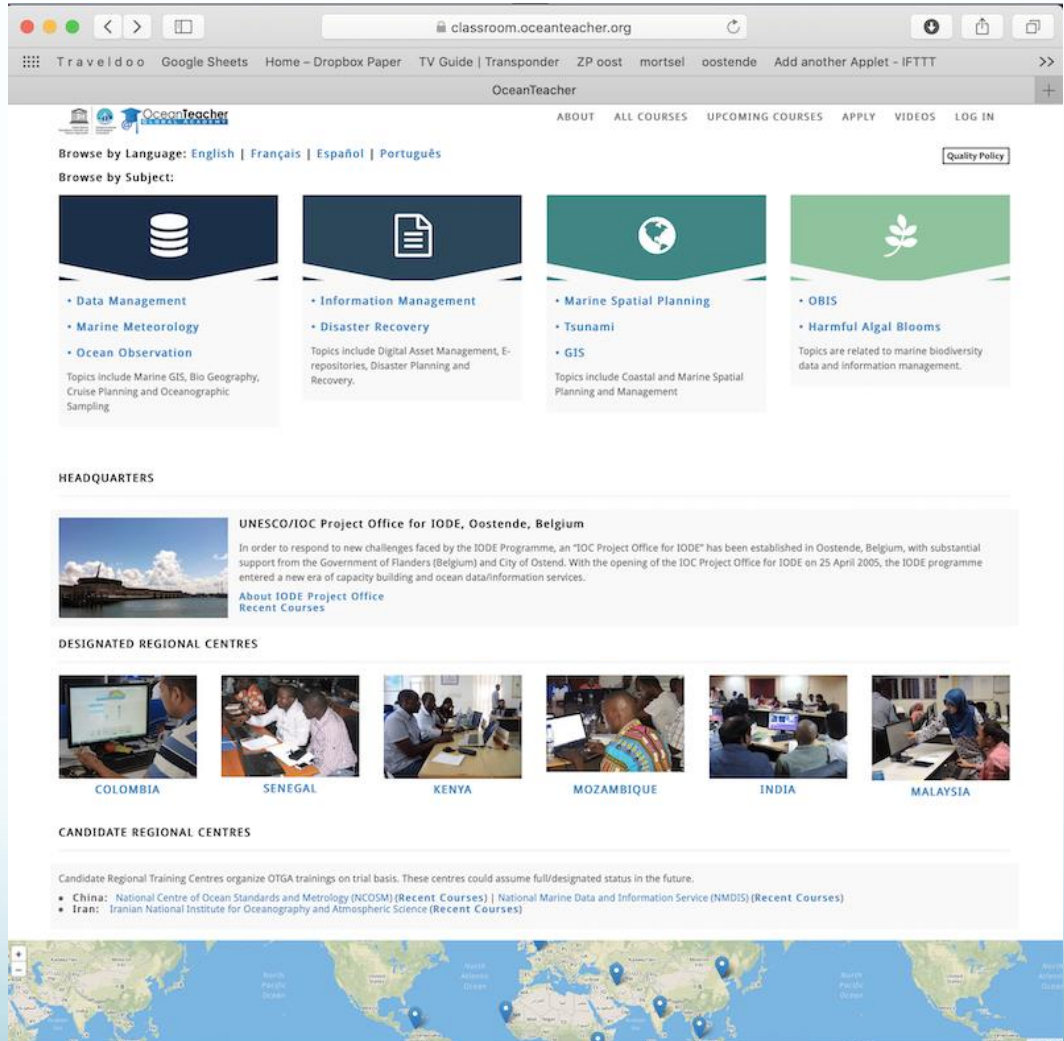
Training?

OceanTeacher Global Academy

Global network of
Regional Training Centres

Currently focus mainly on
IODE themes but
additional themes and/or
centres can be established

Strong community
involvement needed to
develop courses and
lecture courses



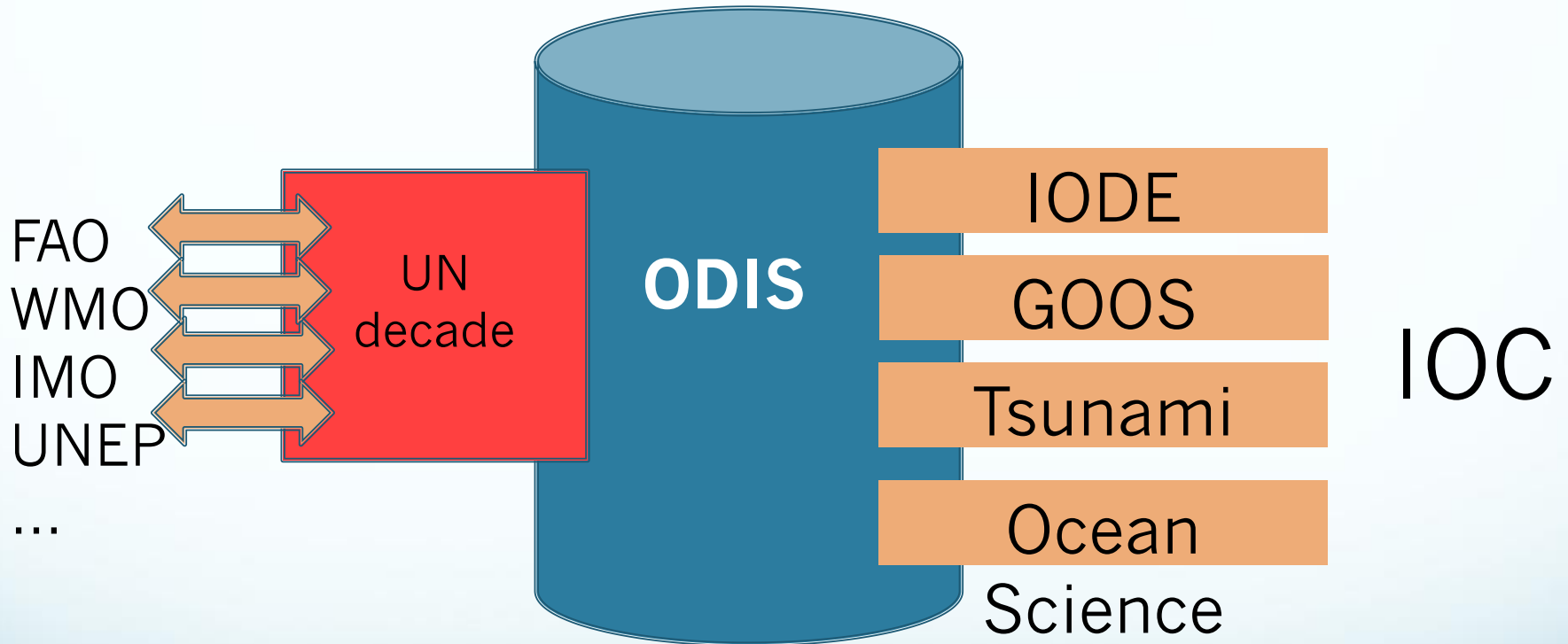
The screenshot shows the OceanTeacher Global Academy website. The header includes the OceanTeacher logo and navigation links: ABOUT, ALL COURSES, UPCOMING COURSES, APPLY, VIDEOS, and LOG IN. Below the header, there are language options (English, Français, Español, Português) and a Quality Policy link. The main content area is divided into four subject categories, each with a list of topics and a brief description:

- Data Management**: Topics include Marine GIS, Bio Geography, Cruise Planning and Oceanographic Sampling.
- Information Management**: Topics include Digital Asset Management, E-repositories, Disaster Planning and Recovery.
- Marine Spatial Planning**: Topics include Tsunami and GIS. Topics include Coastal and Marine Spatial Planning and Management.
- OBIS**: Topics include Harmful Algal Blooms. Topics are related to marine biodiversity data and information management.

Below the subject categories, there is a section for HEADQUARTERS, featuring a photo of the UNESCO/IOC Project Office for IODE, Oostende, Belgium, and a brief description of the office's role. This is followed by a section for DESIGNATED REGIONAL CENTRES, which includes photos and names of centres in Colombia, Senegal, Kenya, Mozambique, India, and Malaysia. The final section is for CANDIDATE REGIONAL CENTRES, listing China (National Centre of Ocean Standards and Metrology (NCOSM) and National Marine Data and Information Service (NMDIS)) and Iran (Iranian National Institute for Oceanography and Atmospheric Science).

UN decade on Ocean Science for Sustainable Development

2021-2030



QUESTIONS?